
ToPIN: A Visual Analysis Tool for Time-anchored Comments in Online Educational Videos

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Abstract

Online videos are widely used to share content for a variety of entertainment, educational and other purposes. To support social interaction, several video-sharing websites - including Ustream, niconico, and Twitch - allow users to post messages while they are watching videos. As users' comments can be sorted according to the timecode of each video, this is known as time-anchored commenting. We propose a novel visualization method, ToPIN, which is able to analyze and categorize the topics and content types of users' time-anchored comments. We have also developed a visualization interface that combines the visualization techniques of ToPIN and ThemeRiver to generate additional valuable insights for analysts seeking to make sense of time-anchored comments. To test the utility of our approach, we visualized time-anchored commenting data from two online course videos and invited the course instructors to evaluate our system.

Author Keywords

Online Educational Video; Comment Analysis and Visualization; Time-anchored Commenting

ACM Classification Keywords

K.3.1. [Computers and Education]: Computer Uses in Education, Distance Learning.

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Introduction

Online educational videos have become an important medium for self-education including Massive Open Online Courses (MOOCs), Open Course Ware (OCW). A big challenge faced by online courses is that, in the absence of face-to-face communication, it can be difficult for instructors to gain access to learners' feedback that might help them improve their courses. Though it has been recommended that current online course platforms add communicative features such as discussion forums, as well as monitoring/analysis tools including metrics tabs, and Khan Academy's Coach monitoring system, most such platforms still provide only limited sets of summarizations of basic, non-content features for the purposes of monitoring and analysis [11, 13]. The question of how to effectively acquire and display learners' content-oriented qualitative information therefore poses a key challenge for people and organizations that offer online courses.

We set out to develop an efficient visual analytic system capable of dealing with time-anchored comments, in which learners' brief written observations about a video are sorted according to its time-code [8]. We believe time-anchored commenting could also provide an important channel whereby instructors can access learners' feedback and observe their learning status, both quantitatively (timing, counts) and qualitatively (emotion, content). Because time-anchored comments are temporal and usually short and thematically diffuse, our proposed visualization method, ToPIN, offers a qualitative view of the detailed contents of comments, such as the type and time of individual comments as well as topic clustering and evolution. We also utilize ThemeRiver [3] to provide an alternative quantitative view focusing on the variations in the pool of time-anchored comments. The integration of ToPIN and ThemeRiver supports global-to-local, evolutionary, and multi-facet analyses that enable the ana-

lysts to visually explore the interactions among learners as well as between learners and the video lecture.

To evaluate our system as an application in the context of online education, we conducted an evaluative study for course instructors. Two instructors who had produced the content of the educational videos were asked to use our system to analyze learners' responses to their lectures. Both instructors reported very positive experiences with using our visualization.

Related Work

Topic Visualization. Some previous approaches display topics as word clouds or text-based visual descriptions related to the frequency or timestamp of topics [6, 7]. There are also methods that focus on visualizing the trends of topics: ThemeRiver [3] and Textflow [1] use a flowing-river metaphor to analyze thematic variation and the merging and splitting of relationships between topics. TopicFlow [9] aligns similar topics across time slices and visualizes the emergence, convergence, and divergence of topics. However, none of them can be directly applied to the evolution of topics in time-anchored comments, because these comments are usually short and in some cases incoherent, as compared to the types of data that these five systems were designed to deal with.

Conversation Visualization focuses on identifying the temporal or structural relationship between conversations, and/or using text analysis to explore the insights generated by conversations. For example, Conversation Thumbnails [14] visualizes the thread structure of conversations using tree-visualization techniques. Opinion Space [2] visualizes differences in users' opinions by projecting them onto a two-dimensional map. ConVis [4] displays conversations on blogs with a thread overview that shows structure and

conversational sentiments, and a facet overview covering topics and authors. there is an obvious difference that we analyze and visualize the multiple facets of comments from a temporal rather than a structural perspective.

Online-video comment visualization. PolemicTweet [5] provides an overview of the tags (e.g., agreement, disagreement, question) and activity spikes in audiences' annotation of online videos. Conference Monitor (CM) [12] is a real-time, web-based tweet visualization dashboard, which monitors back-channel conversation during academic conferences. As PolemicTweet and CM were not developed for use with online educational videos, they cannot provide deeper insights related to learning experiences, such as the emotional valence and course-relevance of comments. Their success in their own spheres inspired us to develop a visualization system for online educational videos though.

Comment Analysis

To gain an understanding of different aspects of comments, we develop several analysis components by adopting text-processing and machine learning techniques.

Emotion analysis. As learners often express their feelings about these aspects of their learning experiences through time-anchored commenting, our system analyzes and categorizes it into three emotional types - positive, neutral, and negative.

Course-relevance analysis. Analyzing the degree to which comments are relevant to the course is important for instructors to understand learners' learning situation, thus our system classifies comments into two categories, course-related and course-unrelated.

Content analysis. The content of learners' comments is diverse and often complex. Effective classification of content

type must therefore be relatively elaborate. Thus, we have adopted instructors' recommendations by following an existing scheme [10] that divides the content of comments into six types: general conversation, notes, opinion, questions, complaints, and compliments.

Topic analysis and clustering. Topics summarized from learners' comments should provide a good overview of such comments, which may reflect the course content, or social activities that interest learners more.

We preprocessed comments using Chinese text-segmentation module, stop-word filtering and part-of-speech (POS) tag for each word, then adopted supervised learning to classify the emotion, relevance, and content of comments. To avoid using a large amount of training data, which would usually be manually labeled, we chose a naïve Bayes classifier to classify comments. Each type of analysis is formulated as a two-class classification problem, i.e., positive vs. negative emotion, and a specific type vs. the other types in content analysis of comments. For topic analysis, we adopted k-means clustering to group them and measure cosine similarity to as the distance between each comment sentence.

Visualization Method and Implementation

Visual Encoding of ToPIN

ToPIN depicts the evolution of topics in time-anchored comments as boxes scattered in a two-dimensional space. The x-axis is the time axis and the y-axis displays the values of a user-chosen attribute (Fig. 1). Each box is attached to the time axis by a series of curved lines of various colors, each of which denotes a comment, and all connected to the time axis. The intersection-point of each curved line with the time-axis indicates the timecode in the video at which the comment was left. The density of the curved lines therefore indicates the distribution of comments along the timeline,

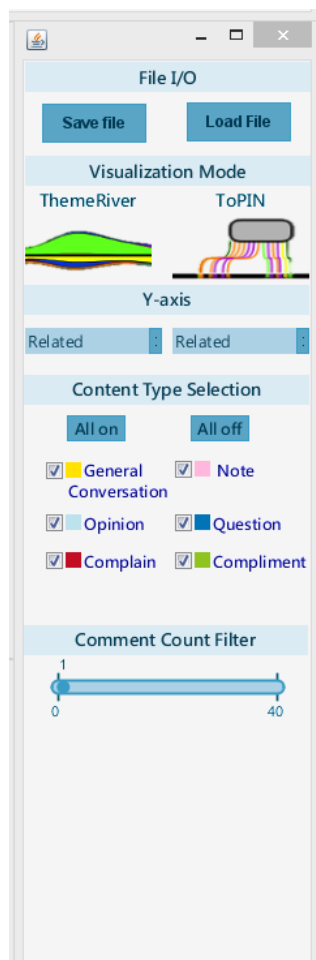


Figure 2: The control panel.

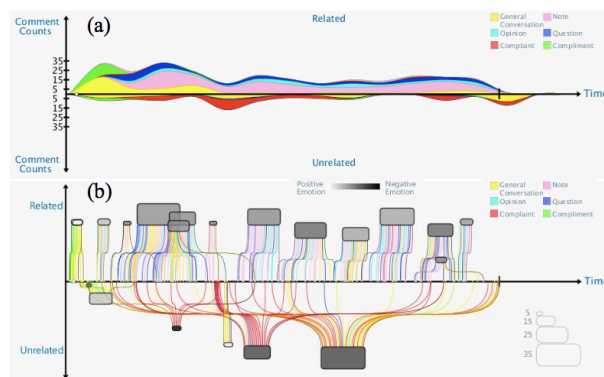


Figure 1: (a) ThemeRiver and (b) ToPIN visualizations of learners' time-anchored comments on a philosophy course.

and their colors represent the comment types, e.g., yellow for general conversation.

The popularity of a topic (measured by the number of comments that mention it) is visually represented by box area. In addition, the shade of a topic box represents the average emotional values of all comments associated with the box, with lighter shades corresponding to more positive emotional values. In other words, ToPIN allows its users to efficiently explore the topics of learners' discussions about a course, along with the relative importance and attributes of each topic.

Interface

We developed an interactive visual analytical system that integrates ToPIN and ThemeRiver. ToPIN shows the topics summarized from individual comments as well as the emotional tone and course-relevance of each topic. Although ThemeRiver [3] is suitable for presenting the data in terms of long-term continuous variations, time-anchored

comments on course videos are, by their nature, distributed unevenly and intermittently in the video timeline. Thus, we utilize ThemeRiver to depict and provide quantitative analysis of the variations in comment types. ThemeRiver and ToPIN complement each other and offer users an effective means of exploring learners' feedback regarding different parts of a course video.

We used *Processing* (<https://processing.org/>) to implement our system. Its interface consists of four views: ThemeRiver, ToPIN, Course Video, and Comment. The control panel (Fig. 2) integrates several user-interface functions described below.

- **Highlighting.** When a user clicks a topic box in the ToPIN view, the topic box and all comment curves connected to that box are highlighted and the keywords describing the topic are listed in a pop-up box (Fig. 3). In addition, the Comment view lists only those comments associated with the highlighted topic. Similarly, in the ThemeRiver view, clicking on a flow will highlight that flow, which represents a specific type of comments. The highlighting function simultaneously updates the ToPIN, ThemeRiver, and Comment views.
- **Display filters.** A user can choose whether s/he sees all, or only certain types of comments by checking some of the six boxes labeled Content Type Selection. This could help users to find out which part of a course video prompted the most questions, or more notes on the important part of a lecture. The ToPIN, ThemeRiver, and Comment views are updated simultaneously when the user makes changes to Content Type Selection. We also designed a Comment Count Filter, operated via a slider bar, which allows users to screen out less popular topics.

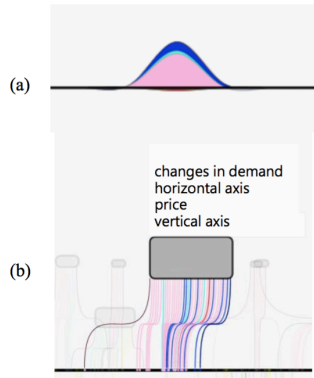


Figure 3: Clicking a topic box in ToPin or a flow in ThemeRiver, the topic box and flow are highlighted and the keywords are listing in a pop-up box.

- Synchronous time bars. The time bars in the course video, ToPIN, ThemeRiver, and Comment views are synchronously updated. When the user scrolls a time bar in any of these views, the time bars in the other views will be updated to the same time instant. This ensures that users are able to accurately explore the contextual relations among comments, course video, and visualizations.
- Plot selection. In the control panel, the user can choose to see only ThemeRiver, only ToPIN, or both of them by clicking a visualization mode. In addition, the user can plot emotional valence or course relevance on the y-axis. If course relevance is chosen, a topic box located in the upper part of the screen will indicate the relevance level of its contents to the course content is high.

Evaluation

Data Collection and Classifier Testing

The time-anchored comments were collected in a previous study [8]. Three online courses were selected from the OCW and Coursera of major universities in Taiwan. One video clip with a length of approximately 15 minutes was selected from each course. The number of comments collected from the Neural Science, Economics, and Philosophy courses were 430, 420, and 427, respectively. We then convened a focus group consisting of two researchers who coded the emotional tone, course relevance, and content type of each comment, and grouped comments into topic clusters. When the two human-coding results reached 90% similarity, one of them was adopted as the ground truth for comment classification and analysis. For each course, one-third of the comments were used for training and the others for testing. The average classification rates for the three courses at the test stage ranged from 72% to 90%.

Participants and Task Design

We invited the instructors of the Neural Science (P1) and Economics (P2) courses to evaluate our system. We focused on testing the overall usability and usefulness of our system and how well it assisted the instructors in exploring learners' feedback. In the first task, the instructors were asked to observe the trends in learners' participation in their courses. In the second task, the instructors were asked to tell us which course content induced heated discussion among the learners. In the third task, we allowed the instructors use our system to freely explore learners' comments.

Procedure

We first introduced our visualization system to each instructor and conducted pre-interviews with them to understand their experience of teaching on online courses and what information they wanted to obtain from learners' feedback. After each pre-interview, we demonstrated our interface and gave instructors practice task using the comments on the philosophy course, which were not subsequently used in the study itself. Neither of the instructors failed to complete the practice task. Once the main study tasks were completed, we interviewed each instructor again to learn about their user experiences and what they had found out about the students' course experiences.

Results and Discussion

The current focus of evaluation is on the effects of the visualisation design itself, rather than the influence of the performance of automatic comment classification, thus manually classified comments were fed into our system in the current study. Once we confirm the usefulness of the visualisation design, we may test the system with machine classification of comments for large scale deployment. In task one, both participants tended to first observe the flow of the en-

tire ThemeRiver to see where the most or fewest comments had been left. They then selected different content types to observe the flow of each. Both P1 and P2 concluded that the learners were engaged by the course videos and had heated discussion on diverse topics including both course-related comments (e.g., notes) and course-unrelated ones (e.g., general conversation). On the other hand, in task two, both P1 and P2 chose to set a higher threshold in the comment count filter to screen out less popular topics. To locate the most active discussions, they both selected opinion content type in the control panel and used ThemeRiver to observe the flow of opinion comments.

In free exploring task, P1 mainly used the ToPIN view and the comment-count filter to identify major topics. He browsed all the topics containing question-type comments and then he found some questions from learners that revealed their misunderstandings of terminologies and definitions, as well as confusion that had been caused by his unclear pronunciation. P2 first viewed the peaks of each flow in ThemeRiver. Browsing the flow of compliments, P2 found a high peak at the end of the video. He then switched to ToPIN, and found there were some topic boxes around the time-point of the peak containing many compliments about his interesting way of teaching and the excellent examples he had used to illustrate the difference between superior and inferior goods.

The observations in this study showed that the instructors used ThemeRiver to assess the overall situation of learning with the course video (i.e., the global pattern), and then used ToPIN to observe the details of learners' comments (i.e., the local details). The instructors also indicated that the flows of different colors in ThemeRiver were very suitable for illustrating student-commenting trends. They also reported that the topic keywords that appeared when a topic box was clicked helped them to draw conclusions

about the learners' comments efficiently; that the size of the boxes indicated the number of comments intuitively; and that the curved lines presented the distribution of the comments clearly.

Both instructors said that one of the greatest strengths of our visualization was that it showed temporal patterns of topics, comments, and course video integratively.

"Because the topic keywords of the learners' comments are displayed, I can easily retrieve the corresponding course topics." (P1)

"By using the system, I can identify the reasons that the negative emotions or some specific responses of learners were generated. After I conclude the reason, I can come up with some solutions to prevent the problem from happening again." (P2)

This verified that our system can help instructors to understand which content interests learners the most and least (among other matters) and locate special events or responses from learners' comments through visual analysis.

Conclusion and Future Work

This paper presents a comment analysis and visualization system for online education videos. The evaluation studies participated by the course instructors validated the efficacy of our system. Their findings further suggested that time-anchored commenting can be a very useful channel for monitoring online learners' feedback and with the assistance of our visualization technique, the course instructors can gain insights into students' learning status and quality of learning. In the future, we would like to apply our system to the comments from actual online learners and generalize our findings at a larger scale study.

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